India's Rise in Open Source Software

An ecosystem of opportunities

Introductions

Samir Kamerkar
Engineering Director
Canonical

Computer Engineer by education,
Java developer by experience,
Problem solver by nature,
Coach at heart.

linkedin.com/in/samirkamerkar/
X - @samirkam

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If I had a rupee for everytime someone told me that Java is dead over the last 25 versions...

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According to the tiobe index **Python, C/C++, Java, C#, Go, Javascript** are amongst the most popular programming languages.

TIOBE

1	1		Python
2	4	^	C
3	3		Java
4	2	v	C++
5	5		C#

… and all of them are being developed by open source communities

https://github.com/python

https://github.com/openjdk/

https://gcc.gnu.org/git.html

https://github.com/dotnet

https://github.com/golang/go

https://github.com/nodejs

https://github.com/rust-lang

India: Fastest-Growing Developer Community

17M+ developers on GitHub (2024)

28-31% annual growth rate

2nd highest in open-source contributions

2nd in AI/ML repository contributions (79% YoY growth)

Only about 10% of them are employed...

Rest are students and hobbyists!

But who pays for open source development?

Yes!

Even MNCs do...

- IBM + Red Hat
  - Java
  - node.js
  - golang
- Microsoft
  - .Net
  - Java
- Google
  - golang
- Amazon
  - Rust
- Canonical
  - All of them

The number of open source development jobs has increased exponentially

>900 jobs in India on Linked-in

> 500 jobs remote on indeed

>2500 jobs on naukri.com

Even if we only consider 50% of this number as genuine good opportunities, that is >2000 jobs

> 100

Language and runtimes jobs at 5% of genuine open source jobs posted at any given time

Who can get these jobs?

Practically, every role in SDLC

Application Developers

Devops Engineers

Software Egnineers in Test / QA

Technical Authors

<Language> application developers

If you currently have skills in a programming language, you can contribute to that language or runtime development in the communities

Made by FRE

Devops Engineer

If you have skills in build and test automation, you will be in demand at Open source projects to build and maintain their CI/CD pipelines

QA / Software engineer in Test

Everything that is built, needs to be tested, including open source projects. So if you are SE in Test, you can become the gatekeeper of quality in the project.

Technical Writer

Documentation is a key driver in adoption by users as well as participation by contributors, in an open source project. So if you are a technical writer, you will be sought after.

So how do you get started?

Open source is welcoming

All communities have a getting started guide or new contributors page

Most have a ‘helpwanted’ issues list

Many have a ‘goodfirstissue’ list as well

Reading code is the best way to learn

Contribution guides to get started

- Python
- Rust
- node.js
- golang
- .Net
- OpenJDK
- Ubuntu

Some Java Projects in Ubuntu

OpenSSL-FIPS-Java

OpenJDK-CRaC

devpack-for-spring

Java-rockcraft-plugins

From contributor to committer

Open source communities provide excellent growth path to eminence, if you -

Contribute often - no contribution is small

Contribute quality - clean code is your currency

Contribute value - work on priority issues

Find a coach / mentor

Having an experienced navigator helps tremendously on the open source journey.

- Ask and you will find someone who is willing
- Do your homework before going to the coach
- Don’t expect to get help served on a platter
- Respect your coach's time and calendar.
- Be clear and precise about what you need help with
- Seek help sooner rather than later
- Identify the time sensitivity of your need and schedule accordingly

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It's surprising that for the scale of talent, there aren't many well known Open source thought leaders from India

There are only a handful open source community leaders from India

So the scope to standout is tremendous!

Questions?